

The Distribution of Agency in AI Systems

Agency in artificial intelligence systems operates across multiple interconnected layers, each with distinct decision-making power and responsibility deflection mechanisms.

Developers

- Choose algorithms and architectures
- Select training data
- Define metrics and objectives
- Implement guardrails (or don't)

"I just optimised for accuracy"

Companies

- Decide what problems to solve
- Choose deployment contexts
- Set policies and monitoring
- Determine resources allocated

"We're just meeting market demand"

Users

- Decide whether and how to use
- Interpret and apply outputs
- Verify (or don't verify) results
- Provide feedback

"I just used what the system recommended"

Regulators & Society

- Set requirements and standards
- Enforce (or fail to enforce)
- Allocate liability
- Shape incentives

"We're still figuring out the rules"

When Agency Attribution Goes Wrong

When we attribute agency to AI, human responsibility disappears

Hiding Behind "The AI"

"The AI decided not to hire you"

"The algorithm determined your creditworthiness"

"The system flagged your content"

Reality: Humans designed the system, chose the data, defined the objective, deployed it, and continue to use it

Responsibility Disappears

Users say: "The AI told me to do it"

Designers say: "I just built what was requested"

Companies say: "The system made an error"

Regulators say: "It's too complex to understand"

Result: No one takes responsibility. Harm occurs without accountability.

Three Ways Responsibility Disappears

Many Hands Problem

When everyone is partially responsible, no one accepts full responsibility. "Not my job" — everyone points to someone else, creating diffusion of responsibility. This mirrors the classic Kitty Genovese bystander effect, now manifested in technological systems.

"Computer Says No" Phenomenon

Decisions are attributed to "the system" rather than human judgement. This creates psychological distance from outcomes and obscures the human choices that built and configured the system. "The algorithm decided, not me."

The Complexity Excuse

Systems become too complex for anyone to fully understand. "We can't explain how it works" transforms opacity into a shield against accountability. "It's a black box—even we don't know why."

❏ These are not inevitable technical limitations. These represent human choices about how we structure systems and accountability frameworks. We could design differently, maintaining clear accountability – but this requires deliberate, sustained effort.

Why We Attribute Agency to AI

Humans naturally attribute agency to systems that exhibit certain characteristics, even when those systems have no real understanding, intentions, or consciousness.

Language Fluency

When systems produce coherent, contextually appropriate language, we instinctively treat them as understanding. Our social cognition is triggered by linguistic competence, even when it's purely statistical pattern matching

Anthropomorphic Design

Chatbots with names, first-person language ("I think," "I believe"), avatars, and human-like responses encourage us to treat AI as an agent with mental states. This is often intentional design, not accident.

Opacity & Complexity

When we can't understand how a system works, we fill the gap by attributing it agency ("the algorithm decided," "the AI wants") rather than acknowledging our own ignorance about the mechanism

The ELIZA Effect (1966)

What ELIZA Was

A simple chatbot created by Joseph Weizenbaum at MIT that simulated a psychotherapist using basic pattern matching and scripted responses

Example exchange:

User: "I'm worried about my mother"

ELIZA: "Tell me more about your family"

The mechanism: Simple keyword detection + pre-written response templates. No understanding whatsoever.

What Happened

Users attributed deep understanding and empathy to ELIZA – even when told it was just a program

Weizenbaum's secretary asked him to leave the room so she could have a "private" conversation with ELIZA

Some psychiatrists proposed ELIZA could automate therapy

Surface-level linguistic competence triggers attribution of agency, understanding, and care

Instrumental vs. Genuine Agency

A critical distinction for understanding AI ethics:

Instrumental Agency

"Acting as if it has purposes"

- Appears goal-directed
- Optimizes for objectives
- Adapts behavior
- Produces coherent outputs
- Described "as if" intentional

✓ AI systems have this

Genuine Agency

"Actually having purposes"

- Self-determined goals
- Conscious intentionality
- Understanding of stakes
- Moral responsibility
- Autonomy and freedom

✗ AI systems (currently) lack this

Intelligence as a Commodity

For the first time in history, cognitive capabilities can be purchased at scale like electricity or computing power

What This Means

Pay for Intelligence, Not People

Companies can purchase cognitive capabilities (writing, analysis, coding, translation) without hiring humans—API calls instead of salaries

Democratization and Concentration

Anyone with internet access can use sophisticated AI (democratizing)—but a few companies control the infrastructure (concentrating power)

Economics Fundamentally Shift

Marginal cost of cognitive work approaches zero—one AI model serves millions. Creates winner-take-all dynamics and new forms of inequality

Two Industrial Revolutions

First: Muscle Power on Demand

18th-19th century: Steam engines, machines, factories

What changed: Physical labor became available as a commodity—you could buy muscle power

Impact on work: Manual labor displaced, new jobs emerged, massive economic restructuring

Power shifted: Those who owned factories and machines accumulated wealth

Yet today: People still exercise, play sports, enjoy physical activity—even though machines are stronger

Second: Intelligence on Demand

21st century: LLMs, AI systems, automation platforms

What's changing: Cognitive labor becoming available as a commodity—you can buy intelligence

Impact on work: Knowledge work displaced, new jobs emerging, massive economic restructuring

Power shifting: Those who own AI platforms and compute are accumulating wealth

The future: People will still think, solve problems, enjoy intellectual challenges—even though AI is faster

Impact on Jobs and Work

Most Vulnerable: Routine Cognitive Work

Examples: Data entry, basic coding, document summarization, translation, customer service scripts, basic legal research, initial medical diagnoses

Why: Pattern-based, large training data available, clear success metrics

Augmented: Complex Knowledge Work

Examples: Software engineering, scientific research, strategic planning, creative design, teaching, journalism

Why: AI assists but human judgment, creativity, and contextual understanding remain essential. **Will this lead to augmentation or deskilling?**

More Protected: Physical & Social Work

Examples: Healthcare (hands-on), trades (plumbing, electrical), childcare, eldercare, therapy, arts performance

Why: Requires physical presence, human connection, trust, or real-world manipulation that AI cannot (yet) replicate

Power Concentration

AI amplifies existing power dynamics. Those with resources gain disproportionately

Compute & Infrastructure

Training frontier models requires billions in investment. Only a handful of companies can afford this scale.

Data & Network Effects

More users → more data → better models → more users. Winner-take-most dynamics.

Productivity Gains

Workers become more productive, but who captures the value? Historically: capital owners.

What Endures?

AI Will Do Everything

"AI will replace all knowledge workers" • "Human intelligence is obsolete" • "Nobody will write code anymore"

Human judgement and control

Some tasks automated, others augmented. Many require human judgment. The question is not "will AI replace you" but "how will you work with AI?"

The Sports Analogy: Using Your Mind Despite AI

Even though machines are far stronger, people still run marathons, climb mountains, do crafts by hand, and value physical fitness.

Similarly: People will continue to think, learn, and create—not because it's most efficient, but because *using our intelligence is intrinsically valuable*.

The Sports Analogy

Physical Activity

Machines are **far stronger** than humans, yet people still:

- Play sports for enjoyment
- Exercise for health and challenge
- Value physical accomplishment
- Find meaning in using their bodies

Cognitive Activity

AI is **far faster** than humans, yet people will still:

- Think for enjoyment
- Solve problems for satisfaction
- Value intellectual accomplishment
- Find meaning in using their minds

Just because AI can do something faster doesn't mean humans won't choose to do it themselves